Science practice booklet: Grade 10 expectations

Grade 10 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 10 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: HS-LS2-1, HS-LS2-2, HS-LS2-4, HS-LS2-6, HS-LS3-1, HS-LS3-2, HS-LS3-3, HS-LS4-3, HS-LS4-5

A worked example

In a mangrove ecosystem, algae capture 20000 units of energy from sunlight. Small fish eat the algae, crabs eat the small fish, and herons eat the crabs. About 10 percent of the energy is passed on at each step. How much energy reaches the herons?

1. Algae are the producers. Small fish are the first consumers, crabs the second consumers and herons the third consumers.
2. Algae to small fish: 10 percent of 20000 = 2000 units.
3. Small fish to crabs: 10 percent of 2000 = 200 units.
4. Crabs to herons: 10 percent of 200 = 20 units.
5. The rest of the energy at each step is used for life processes or given off as heat.

About 20 units of energy reach the herons.

Questions

1. What is meant by the carrying capacity of an ecosystem?

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2. A reserve starts with 100 Arabian oryx. The numbers rise quickly for several years, then stay close to 600 for a long time. What does the number 600 show?

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3. Name two limiting factors that can keep a desert animal population from growing, and say which are living (biotic) and which are non-living (abiotic).

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4. Producers in a lagoon capture 50000 units of energy. About 10 percent is passed on at each step. How much energy reaches the second consumers?

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5. Describe how carbon moves between the air and living things. Name the two key processes.

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6. Why does a mangrove forest with many different species usually recover better after a storm than a place with only one or two species?

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7. Genes come in different versions called alleles. In a garden plant, red flowers (R) are dominant and white flowers (r) are recessive. Two plants with the genotype Rr are crossed. What fraction of the offspring is expected to have white flowers?

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8. Give two ways in which variation between individuals of the same species can arise.

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9. Some desert plants have thick, waxy stems and deep roots. Explain how these traits help the plants survive.

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10. Arabian oryx have pale coats that reflect sunlight and hooves that spread weight on soft sand. Explain how natural selection could produce these traits in a group over many generations.

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11. Explain why one individual cannot change its inherited traits to suit a new environment, but a population can change over generations.

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12. Building along a coast removes mangrove trees, and the number of fish and crabs there falls. Explain two ways in which human activity can reduce biodiversity, and suggest one action that helps.

Section 2: Earth and space

Standards practised: HS-ESS1-1, HS-ESS1-3, HS-ESS1-4, HS-PS2-4

A worked example

Light travels at about 300 000 km per second. Mars is on average about 228 000 000 km from the Sun. How long does sunlight take to reach Mars?

1. Time = distance ÷ speed.
2. 228 000 000 ÷ 300 000 = 760 seconds.
3. 760 ÷ 60 = 12 with 40 left over, so it is 12 minutes 40 seconds.

About 760 seconds, which is about 12 minutes 40 seconds.

Questions

1. What nuclear process produces the Sun's energy?

2. The Sun stays about the same size for billions of years. Which two things are in balance inside it?

3. Put these stages of a star like the Sun in order: white dwarf, nebula (cloud of gas and dust), main sequence star, red giant.

4. How does the end of a star with a very large mass differ from the end of the Sun?

5. Elements such as carbon and oxygen are made inside stars. How can this matter later become part of new planets?

6. Light from Mars is sent to Earth at a time when the two planets are 60 000 000 km apart. Light travels at 300 000 km per second. How long does the light take?

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7. A light year is a unit of distance. What does it measure, and why is it useful?

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8. Scientists split starlight into a spectrum and see dark lines at certain colours. What can the lines tell them about the star?

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9. Two stars are the same distance away. One looks blue-white and the other looks red. Which one has the hotter surface?

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10. The force of gravity between two objects gets smaller as they move apart. If a satellite moves to twice its distance from the centre of Earth, by what factor does the pull of Earth change?

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11. Planets closer to the Sun complete an orbit in less time than planets farther away. Mercury takes 88 days and Earth takes 365 days. Explain why this is expected.

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12. The Hope Probe travels in an elliptical (oval) orbit around Mars. Using gravity, explain why the probe moves fastest when it is nearest to Mars.

Section 3: Matter, forces and energy

Standards practised: HS-PS2-1, HS-PS2-2, HS-PS2-3, HS-PS2-4

A worked example

A trolley of mass 1000 kg moves at 20 m per second and joins with a parked trolley of mass 1000 kg so that they move together. What is their speed after joining? (Ignore friction.)

1. Momentum = mass × velocity.
2. Momentum before = 1000 × 20 + 1000 × 0 = 20000 kg m/s.
3. Momentum is conserved, so after joining it is still 20000 kg m/s.
4. Total mass = 2000 kg, so velocity = 20000 ÷ 2000 = 10 m per second.

10 m per second.

Questions

1. State Newton's first law of motion and give an everyday example.

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2. A scooter and its rider have a total mass of 80 kg. An unbalanced force of 240 N pushes them forward. What is their acceleration? Use force = mass × acceleration.

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3. The same force is applied to a 2 kg trolley and a 4 kg trolley. Which has the larger acceleration, and how many times larger?

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4. What is the difference between mass and weight? Find the weight of a 50 kg astronaut where the gravitational field strength is 3.7 N per kg (as on Mars). Use weight = mass × field strength.

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5. A rocket pushes hot gas out of its engine downwards. Use Newton's third law to explain why the rocket moves upwards.

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6. A car of mass 1500 kg travels at 20 m per second. What is its momentum? Use momentum = mass × velocity.

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7. A cart of mass 2 kg moves at 3 m per second and hits a cart of mass 1 kg that is at rest. They stick together. What is their speed afterwards?

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8. A football of mass 0.5 kg is kicked from rest and leaves the foot at 20 m per second. The foot is in contact with the ball for 0.02 seconds. Use force = change in momentum ÷ time to find the average force.

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9. Explain how an airbag and a crumple zone reduce the force on people in a car crash.

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10. A stone is dropped from rest and falls for 3 seconds. Ignore air resistance and use g = 10 m per second squared. Find its speed after 3 seconds and the distance it has fallen.

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11. Two objects attract each other with a gravitational force. What happens to the force if the mass of one object is doubled, and what happens if the distance between them is doubled?

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12. Two small balloons are rubbed with cloth so that both gain the same kind of charge. What happens when they are brought near each other? By what factor does the electric force change if the distance between two charges is halved?

Answer guide for parents

Section 1: Living things

1. The largest number of individuals of one species that the ecosystem can support over time, given its food, water, space and other resources.
2. It is about the carrying capacity of the reserve. The food, water and space can support roughly 600 oryx.
3. Examples: food supply or predators (biotic); water supply or high temperature (abiotic).
4. 50000 × 0.1 = 5000 units reach the first consumers, then 5000 × 0.1 = 500 units reach the second consumers.
5. Plants take carbon dioxide from the air and build it into sugars by photosynthesis. Animals eat the plants. Plants, animals and decomposers release carbon dioxide back into the air by cellular respiration.
6. Greater biodiversity means more food sources, more roles filled and more links in the food web, so if one species is harmed others can keep the ecosystem working.
7. One quarter (25 percent). The offspring are RR, Rr, Rr and rr in the ratio 1 : 2 : 1, and only rr gives white flowers.
8. Mutations (changes in DNA) and the mixing of genes from two parents in sexual reproduction (new combinations of alleles). Environment can also affect some traits.
9. The waxy surface reduces water loss by evaporation, and deep roots reach water far below the surface. Plants with these traits survive and reproduce more often.
10. Individuals vary. Those with pale coats and wide hooves were slightly more likely to survive heat and move across sand, so they had more young. The helpful traits were inherited, and over many generations they became common in the group.
11. An individual's genes stay the same during its life. In a population, the individuals that happen to have helpful traits leave more offspring, so the share of those traits rises from one generation to the next.
12. Habitat loss (the mangroves are the shelter and nursery for many species) and pollution or over-use of resources reduce the number of species. Helpful actions include protecting and replanting mangroves or creating protected reserves.

Section 2: Earth and space

1. Nuclear fusion: hydrogen nuclei join to form helium nuclei in the Sun's core, and a small amount of mass is changed into a large amount of energy.
2. The inward pull of gravity and the outward pressure produced by the hot gas heated by fusion.
3. Nebula, main sequence star, red giant, white dwarf.
4. A star with a very large mass ends in a huge explosion called a supernova, leaving a neutron star or a black hole. The Sun will end as a white dwarf.
5. When large stars explode, or older stars throw off their outer layers, the elements are spread into space. Later this gas and dust can gather into new stars and planets.
6. 60 000 000 ÷ 300 000 = 200 seconds, which is 3 minutes 20 seconds.

7. It is the distance light travels in one year (about 9.5 trillion km). It is useful because the distances between stars are far too large to describe sensibly in kilometres.
8. Each element absorbs light at its own particular colours, so the pattern of dark lines shows which elements are in the star's atmosphere.
9. The blue-white star. Hotter objects give out light with a shorter wavelength, so they look bluer.
10. It becomes one quarter as large (the force depends on 1 divided by the distance squared, and 2 × 2 = 4).
11. The Sun's gravity pulls more strongly on nearer planets, so they move faster. They also have a shorter path to travel, so they complete an orbit sooner.
12. Mars's gravity pulls more strongly when the probe is nearer, and as the probe falls towards Mars its speed increases. Farther away the pull is smaller and the probe moves more slowly.

Section 3: Matter, forces and energy

1. An object stays at rest, or moves at a steady speed in a straight line, unless an unbalanced force acts on it. Example: passengers lurch forward when a bus brakes, because their bodies keep moving.
2. Acceleration = 240 ÷ 80 = 3 m per second squared.
3. The 2 kg trolley, by 2 times. Acceleration = force ÷ mass, so half the mass gives double the acceleration.
4. Mass is the amount of matter (in kg) and stays the same everywhere. Weight is the pull of gravity on the mass (in newtons) and depends on where you are. Weight = 50 × 3.7 = 185 N.
5. The rocket pushes the gas down, so the gas pushes the rocket up with an equal force in the opposite direction. These forces act on different objects.
6. 1500 × 20 = 30000 kg m/s
7. Momentum before = 2 × 3 = 6 kg m/s. After, the total mass is 3 kg, so speed = 6 ÷ 3 = 2 m per second.
8. Change in momentum = 0.5 × 20 = 10 kg m/s. Force = 10 ÷ 0.02 = 500 N.
9. They make the car and the passengers take longer to stop. The change in momentum is the same, so a longer stopping time means a smaller average force.
10. Speed = 10 × 3 = 30 m per second. Distance = ½ × 10 × 3 × 3 = 45 m.
11. If one mass is doubled, the force doubles. If the distance is doubled, the force becomes one quarter as large.
12. They repel, because like charges repel. The electric force depends on 1 divided by the distance squared, so halving the distance makes the force 4 times as large.